

LINUX SERVER HARDENING & SECURE CONFIGURATION REPORT

1. INTRODUCTION

Server hardening is the process of securing a Linux server by reducing its attack surface, eliminating unnecessary services, applying patches, and enforcing strong security configurations. This report demonstrates practical Linux hardening steps performed on Ubuntu/Kali Linux systems.

2. INITIAL SYSTEM ASSESSMENT

- Reviewed existing users using: `/etc/passwd`
- Checked sudo privileges using: `sudo -l`
- Identified open ports using: `netstat -tulnp` or `ss -tulnp`
- Listed running services using: `systemctl list-units --type=service`
- Checked installed packages using: `dpkg -l`

3. USER MANAGEMENT & LEAST PRIVILEGE

- Removed unused accounts using: `sudo userdel username`
- Locked inactive accounts using: `sudo usermod -L username`
- Restricted sudo access by editing `/etc/sudoers`
- Applied Principle of Least Privilege – users granted only necessary permissions

4. SSH HARDENING

- Disabled root login by setting: `PermitRootLogin no` in `/etc/ssh/sshd_config`
- Enabled key-based authentication
- Disabled password authentication: `PasswordAuthentication no`
- Changed default SSH port (optional)
- Restarted SSH service using: `sudo systemctl restart ssh`

5. SYSTEM UPDATES & PATCH MANAGEMENT

- Updated package list using: `sudo apt update`
- Upgraded packages using: `sudo apt upgrade`
- Enabled automatic security updates using: `unattended-upgrades`

6. FIREWALL CONFIGURATION

- Installed UFW (Uncomplicated Firewall)
- Default deny incoming traffic: `sudo ufw default deny incoming`
- Allowed SSH: `sudo ufw allow 22/tcp`
- Enabled firewall: `sudo ufw enable`
- Verified rules: `sudo ufw status verbose`

7. DISABLING UNUSED SERVICES

- Identified unnecessary services using `systemctl`
- Stopped services: `sudo systemctl stop service_name`
- Disabled services at boot: `sudo systemctl disable service_name`

8. FILE PERMISSION HARDENING

- Secured `/etc/shadow` permissions (root only)
- Used `chmod` and `chown` appropriately
- Restricted sensitive configuration files
- Checked SUID/SGID files using: `find / -perm -4000 2>/dev/null`

9. LOG MONITORING & AUDITING

- Reviewed authentication logs: `/var/log/auth.log`
- Reviewed system logs: `/var/log/syslog`
- Installed Fail2Ban to prevent brute-force attacks
- Used `journalctl` for `systemd` log analysis

10. LINUX HARDENING CHECKLIST

- ✓ Removed unused users
- ✓ Enforced least privilege
- ✓ Disabled root SSH login
- ✓ Enabled key-based authentication
- ✓ Updated system packages
- ✓ Configured firewall rules
- ✓ Disabled unnecessary services
- ✓ Secured file permissions
- ✓ Enabled log monitoring

11. INTERVIEW QUESTIONS & ANSWERS

1. What is Server Hardening?

Server hardening is the process of securing a server by minimizing vulnerabilities, disabling unnecessary services, and enforcing security policies.

2. Why Disable Root Login?

Disabling root login prevents attackers from directly targeting the most powerful account, reducing the risk of privilege escalation.

3. What is Least Privilege?

Least privilege means giving users only the minimum permissions required to perform their tasks, reducing the potential damage of a compromised user.

4. Purpose of Firewall?

A firewall controls incoming and outgoing network traffic, allowing only authorized communication and blocking unauthorized access.

5. Risks of Unused Services?

Unused services increase attack surface and may contain unpatched vulnerabilities, making the system more susceptible to exploitation.

12. FINAL OUTCOME

Through this task, practical Linux security hardening techniques were implemented. The system is now more secure against brute-force attacks, privilege escalation, unauthorized access, and service exploitation.